

— Time Series III: GLS, GARCH, and VAR Models!



GENERAL ASSEMBLY

Agenda: More TS Content!

1. **Generalized Least Squares (GLS) Models**
 - Time series with additional x-variables
2. **Generalized AR with Conditional Heteroscedasticity (GARCH models)**
 - Time series that jump around too much!
3. **Variable Autoregressive (VAR) Models**
 - Time series of many things at once!



— Generalized Least Squares for Time Series Data

OLS For Time Series

A naive approach to time series modeling would be to use OLS - but we know this is not allowed! The **I** assumption (independence) of our LINE assumptions is violated!

Our data aren't **independent** - one observation is directly dependent on the ones before it chronologically.



The I Assumption

OLS parametrization:

$$y_t = \mathbf{x}_t^T \beta + \varepsilon_t$$

where:

$$\varepsilon_t \sim N(0, \sigma^2)$$

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Implicit assumptions of independence!

No subscript

GLS (Generalized Least Squares)

A solution here is to **allow the residuals to track the dependence from row to row**. Technically, this takes the form of modifying the assumed **covariance structure** of our linear model. In math, this is:



GLS

GLS parametrization:

$$y_t = \mathbf{x}_t^T \beta + W_t$$

where W_t is an $\text{ARMA}(p, q)$ process.

Python

Implementing this in Python is easy! We can simply specify the **exog** argument to statsmodels' ARIMA class:

```
ARIMA(endog=y, exog=X, ...)
```

In time series, instead of independent/dependent, they say:

- **ex**ogenous = *x*-variable
- endogenous = *y*-variable

Leggo!



Generalized Autoregressive Conditional Heteroscedasticity

GARCH Modeling for great Profit!



Historical Context

Traditionally, financial modelers have been frustrated with the classical ARIMA approach to TS modeling. Let P_t be the closing price of a particular stock.

- P_t is always non-stationary
- $X_t = \log P_t$ is typically observed to be a stationary “random walk”
- $Z_t = X_t - X_{t-1}$ is also stationary and *appears* to be white noise.
- However, there is strong evidence that Z_t is ***not independent white noise!***
- A lot of financial TS is devoted to studying the dependence structure of Z_t , which is not visible from the ACF.

The ARCH(p) Process

The autoregression with conditional heteroscedasticity (ARCH) process models a time series Z_t by:

$$Z_t = \sqrt{h_t} \varepsilon_t$$

where

$$\varepsilon_t \sim N(0, 1)$$

and

$$h_t = \alpha_0 + \sum_{i=1}^p \alpha_i Z_{t-i}^2$$

The ARCH(p) Process (simplified a little)

The autoregression with conditional heteroscedasticity (ARCH) process models a time series Z_t by:

$$Z_t \sim N(0, h_t)$$

where

$$h_t = \alpha_0 + \sum_{i=1}^p \alpha_i Z_{t-i}^2$$

The ARCH(p) Process (simplified a little)

Complicated? Yeah... We can condense this a little like this:

$$Z_t \sim N(0, h_t)$$

where

$$h_t = \alpha_0 + \sum_{i=1}^p \alpha_i Z_{t-i}^2$$

This is sort of like an AR(p) process but it's actually the conditional variance of Z_t

The GARCH(p, q) Process

We can further generalize this by including something like an MA(q) term:

$$Z_t \sim N(0, h_t)$$

where

$$h_t = \alpha_0 + \sum_{i=1}^p \alpha_i Z_{t-i}^2 + \sum_{j=1}^q \beta_j h_{t-j}$$

The GARCH(p, q) Process

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AR(p)-likeMA(q)-like

GARCH Time!



— Vector Autoregressive Modeling

Time Series Predictions

Up until now, we've been doing **univariate** time series. That is, predicting **one** response variable.

It's possible to do **multivariate** time series modeling. Similar to **simultaneous regression**, we can have more than one y -variable in a single model!



Case Study: Coffee

Let's say we run a coffee shop and wish to forecast our sales over time. We sell two types of coffee: **hot coffee** and **iced coffee**.

We could set up two separate AR models to predict future sales from past ones, but there's a problem here! Hot coffee and iced coffee sales are likely *related*. I'd rather have hot coffee on a cold day, and iced coffee on a hot day.

We can capture this interdependence with a VAR model!

Coffee Interdependence

today's sales

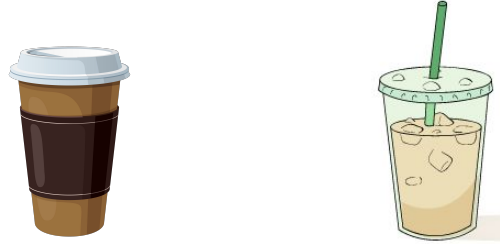


yesterday's sales

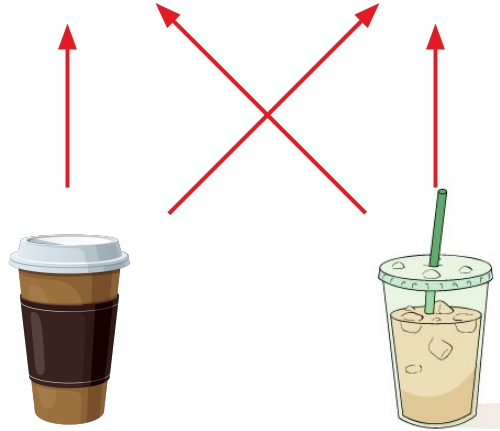


Coffee Interdependence

today's sales



yesterday's sales



VAR Parametrization

$$\begin{aligned}h_t &= \beta_{11} h_{t-1} + \beta_{12} i_{t-1} + \varepsilon_{1t} \\i_t &= \beta_{21} h_{t-1} + \beta_{22} i_{t-1} + \varepsilon_{2t}\end{aligned}$$

VAR Parametrization

$$\begin{aligned} h_t &= \beta_{11} h_{t-1} + \beta_{12} i_{t-1} + \varepsilon_{1t} \\ i_t &= \beta_{21} h_{t-1} + \beta_{22} i_{t-1} + \varepsilon_{2t} \end{aligned}$$

$$\mathbf{y}_t = \begin{bmatrix} h_t \\ i_t \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{bmatrix}$$

$$\varepsilon_t = \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}$$

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$$\mathbf{y}_t = \begin{bmatrix} h_t \\ i_t \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{bmatrix} \quad \varepsilon_t = \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}$$

$$\mathbf{y}_t = \mathbf{B} \mathbf{y}_{t-1} + \varepsilon_t$$

VAR Practice!

